Reg. No. | | | | | | | | | | | [13AE54]

FIFTH SEMESTER B.Tech/B.E EXAMINATION, DECEMEBER–2015

INTRODUCTION SPACE TECHNOLOGY

Time: 3 Hours **Maximum Marks:70**

Instructions: Answer any five full questions.

1. Write a short notes on:

a. Applications of space technology in the commercial domain. **07**

b. Progress in scientific investigations using space platforms. **07**

2. a. How many variables are required fully define the orbit and position of an orbiting body in space? Define these orbital elements and indicate their typical units. **07**

b. Calculate the specific energy of a satellite in the geostationary orbit. Compare it with that of another satellite in a polar sun synchronous orbit of 1000km uniform altitude. Assume radius of earth to be 6378Km and geostationary orbit altitude to be 36000Km. The Gravitational parameters (μ) of earth may be assumed as $3.986 * 10^5$ Km^3/sec^2. **07**

3. a. A black body is radiating maximum energy at a wavelength of 0.483micro meters. Determine its temperature on absolute scale. What is the energy generated by the blackbody per unit area at this temperature over the entire spectral range of its radiation? Assume the value of Stephan – Boltzmann Constant to be $5.67x10^{-8}$ Watts / sq. m $(deg.Kelvin)^4$ **07**

b. What is the nature of radiation belts in the vicinity of earth? Provide an example of the method of protection of satellites passing through those regions around the earth. **07**

4. a. What is the distinction between Specific Impulse and Thrust to Weight ratio with relevance to rocket propulsion? Define these terms and explain their significance. **07**

b. The specific impulse values of propellants used in two different systems with equal thrust performance are 300secs. and 400secs. respectively. If the first system consumes propellants at the rate of 400kg / sec, find the propellant mass rate of consumption of the second system. Also find tis thrust in kilo newton's, and the total impulse for burn time of 50secs. **07**

5. a. Explain working of digital sun sensors and their applications. **07**

b. Provide a comparative analyses of different stabilization schemes in terms of their performance and preferences. **07**

6. a. What is implied by the term 'mission requirements' as it applies to a satellite mission? Provide example of arriving at mission requirements for a communication satellite mission. **07**

b. What are the functions of onboard computer in a satellite? **07**

7. a. Explain the structure of coarse Acquisition code of GPS and indicate accuracies under standard positioning service and differential survey mode. **07**

b. Explain concepts of spatial, spectral and temporal resolutions as applicable to satellite remote sensing systems. What are the considerations in their choices? **07**

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8. a. A satellite transmitter system has a parabolic antenna of 2.5meter diameter and efficiency 2dB and operates at 14 GHz in the down link. The HPA in the transmitter system has power output of 300 watts and the losses in the feed and transmit path are respectively 0.4dB and 0.6dB. Calculate system EIRP. **07**

b. Describe a half wave dipole antenna and depict its radiation pattern with the help of a sketch. What are its performance characteristics? **07**